Run report: run-20261011T1827Z-sidechannel

Eigen-Séta v2: constant-time verification of the decapsulation secret phase (descriptive record)

Study: research/eigen_seta_rust_20261011 · sources: commit 19c641d121097e80f727bcc650a08632e204ba1d

What ran

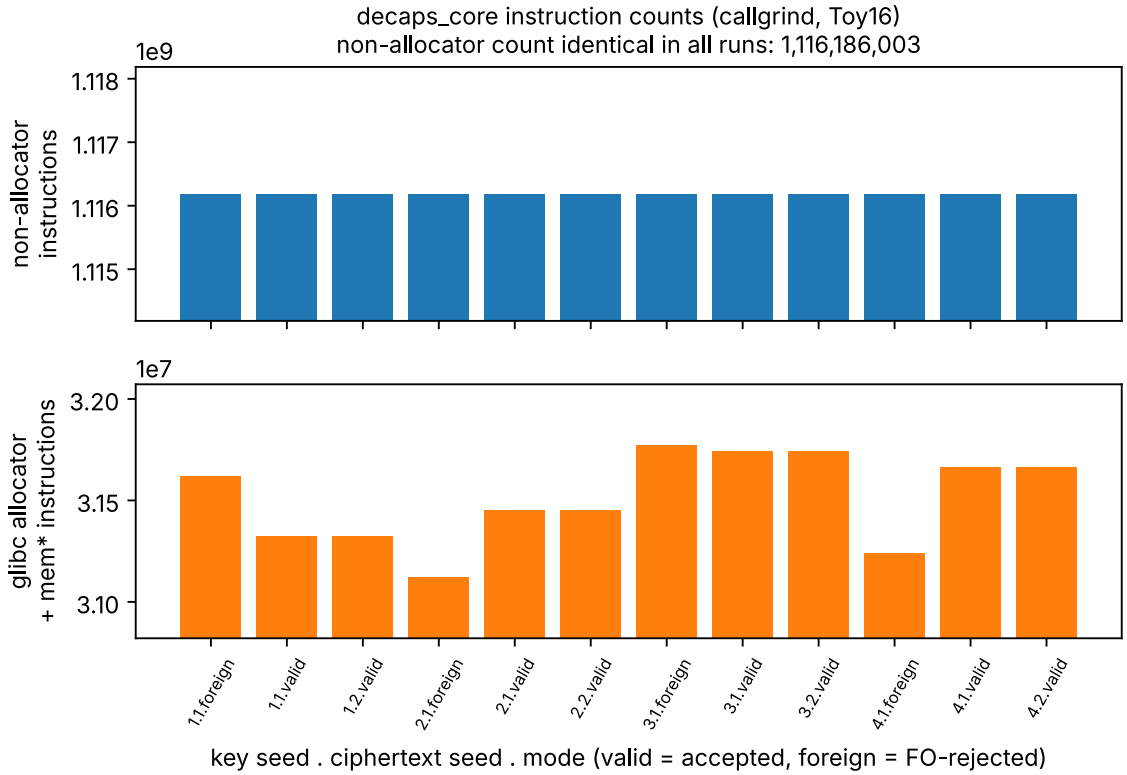
Start (UTC)	2026-10-11T18:27Z, after the benchmark run finished
Sources	Commit 19c641d12. The only uncommitted file in `eigen_seta/` was README.md, which is not compiled.
Build	rustc 1.97.0, release with debuginfo. valgrind-3.22.0 (callgrind), glibc malloc.
Host	Intel Xeon @ 2.80 GHz, 4 logical cores, Linux 6.18.44 x86_64, shared cloud VM
Step 1	`valgrind --tool=callgrind --toggle-collect='*probe_core*' ./ct_probe <key> <ct> <valid|foreign>`. Only `decaps_core` runs inside `probe_core`; the public phase runs outside.
Grid	Key seeds 1–4, each with valid ct 1, valid ct 2 and a foreign ct. A foreign ct is well formed but made for another key, so FO rejects it.
Summary	`ctsummary.sh`: callgrind exclusive counts. `alloc_mem` = glibc malloc, arena and mem* routines. `rest` = total − `alloc_mem`.
Step 2	`./dudect_decaps 1000`: one key. Fixed ct vs. a pool of 16 cts (8 FO-rejected), interleaved at random. Welch t.
Records	`records/ctprobe_summary.txt`, `records/callgrind/*.xz`, `records/probe_stdout/`, `records/dudect_output.txt`, `records/host_and_command.txt`, `records/build.log`

Requested: 12 probe runs and 1 timing test. Completed: 13/13. Failures: none.

Step 1: instructions executed in `probe_core`

key	ciphertext	accepted	total	alloc_mem	rest
1	valid 1	yes	1,147,512,199	31,326,196	1,116,186,003
1	valid 2	yes	1,147,512,199	31,326,196	1,116,186,003
1	foreign	no	1,147,807,708	31,621,705	1,116,186,003
2	valid 1	yes	1,147,639,699	31,453,696	1,116,186,003
2	valid 2	yes	1,147,639,699	31,453,696	1,116,186,003
2	foreign	no	1,147,306,795	31,120,792	1,116,186,003
3	valid 1	yes	1,147,928,378	31,742,375	1,116,186,003
3	valid 2	yes	1,147,928,378	31,742,375	1,116,186,003
3	foreign	no	1,147,958,199	31,772,196	1,116,186,003
4	valid 1	yes	1,147,848,114	31,662,111	1,116,186,003
4	valid 2	yes	1,147,848,114	31,662,111	1,116,186,003
4	foreign	no	1,147,424,282	31,238,279	1,116,186,003

Observation: `rest` takes one value in all 12 runs. `alloc_mem` takes 8 distinct values, and the two valid-ct runs of each key agree exactly.



measured; records: run-20261011T1827Z-sidechannel/records/ctprobe_summary.txt

Figure 1: Instruction counts per run: rest (top) and glibc allocator plus mem* (bottom). Measured; records/ctprobe_summary.txt.

Step 2: timing test (ns per call, 1,000 samples per class)

```
samples/class 1000; median ns: fixed 122309563, random 122318589
t(raw) = +0.75
t(crop <= p50) = +0.84  (n = 501, 500)
t(crop <= p75) = -0.19  (n = 743, 758)
t(crop <= p90) = +0.68  (n = 898, 903)
```

Scope

Toy16 parameters only, one compiler and one target. The check counts instructions, not memory addresses. The timing test uses one key, so it compares ciphertext classes, not keys.